# Engineering in Neuroscience - Assignment #1

EEG Signal Processing in a P300 Paradigm

*BCI Competition III challenge 2004 - Subject A Training Data*

Spring 25 -26

## Under the Supervision of Asst. Prof. ZAFER IŞCAN

| STUDENT INFORMATION | | | | GRADE |
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- **Abstract**

The aim of this assignment was to process EEG data from Subject_A_Train of the BCI Competition III challenge of 2004 dataset and analyze the P300 signal in a brain-computer interface (BCI) speller paradigm. The data was recorded at 240 Hz from 64 EEG channels (electrodes) while the subject focused on characters in a 6x6 letter grid. Rows and columns of the grid flashed randomly, and whenever the target row or column was lit up, the brain produced a characteristic positive response around 300 ms after the stimulus, which is the P300 component. The analysis was done on the Cz channel, which is located at the top center of the scalp and it is known to capture the P300 well. The signal was plotted in the time domain, analyzed in the frequency domain using the FFT, then cleaned using detrending and a 4th order Butterworth bandpass filter (0.5–8 Hz). The power spectrum and log power were also computed. At the end, the average target and non-target responses were compared for the first character epoch, and the P300 effect became clearly visible after filtering, which makes sense as the P300 lives in the low frequency range that the filter keeps.

- **Introduction**

Electroencephalography (EEG) is a method of recording the electrical activity of the brain through electrodes placed on the scalp. It captures the combined electrical signals of millions of neurons firing at the same time, which means the signals are very small and often mixed with a lot of noise. Because of this, signal processing is a really essential part of working with EEG data - without it, it is almost impossible to extract meaningful information from the raw recordings [1].

One of the most well-known and useful signals in EEG research is the P300, which is a type of event-related potentials (ERPs). The P300 is a positive deflection in the EEG signal that appears around 300 ms after a rare or significant stimulus that the person is paying attention to, like a surprise or something was waited for. This makes it really useful for brain-computer interfaces, because you can detect which stimulus the person was focusing on just by looking at where the P300 occurs, without them having to move anything.

To actually analyze the P300, the continuous EEG signal has to be divided into short segments called epochs. This is done using event markers, which are triggers that are recorded at the exact moment each stimulus appears. For each marker, a time window of EEG data is extracted that usually starting slightly before the stimulus and ending a few hundred milliseconds after. Each epoch is then labeled as either a target segment, meaning the stimulus the person was paying attention to, or a non-target segment, meaning everything else. Target segments are expected to contain the P300 component, while non-target segments generally do not. Averaging many target epochs together causes the random noise to cancel out while the P300 survives, which is how the response becomes visible [1].

Before any of this analysis can be done properly, the raw EEG signal needs to be preprocessed. The raw signal contains not just brain activity but also noise from muscle movements, electrode drift, eye blink, power line interference, and other sources [1]. A really common and effective step is bandpass filtering, where only a specific range of frequencies is kept and everything outside is removed. Butterworth filters are widely used for this in EEG because they have a flat frequency response in the passband, meaning they do not distort the frequencies they keep. Since the P300 component is a slow wave that lives roughly in the 0.1–10 Hz range, a bandpass filter that removes high-frequency noise is very helpful for isolating it. Another useful tool is the Fast Fourier Transform (FFT), which converts the signal from the time domain into the frequency domain and shows exactly how much of each frequency is present in the signal. This helps understand what the noise looks like and confirms whether the filter worked correctly.

The BCI Competition III dataset used in this assignment was collected using a P300 speller paradigm, where a subject focused on characters in a 6x6 matrix while rows and columns flashed randomly. The EEG was recorded from 64 channels (electrodes) at a frequency of 240 Hz. The goal of this assignment was to go through the basic steps of EEG signal processing on this data, which are: time domain plotting, frequency analysis using FFT, detrending, bandpass filtering, and finally comparing the averaged target versus non-target responses in order to observe the P300 effect at the Cz electrode. The Cz electrode is located at the very top center of the scalp, channel 11, and it is one of the best positions to capture the P300. The whole analysis was done in Python using Pycharm software.

- **Tools**

  1) Pycharm software for python coding.
  2) Numpy library
  3) Matplotlib
  4) loadmat from scipy.io
  5) Butter, filtfilt, detrend from scipy.signal

- **Results**

**<u>Task 1 - First Trial in Time Domain (Raw)</u>**

The first step was to take the Cz channel signal from the first character epoch and plot the first trial in the time domain. A trial here means the 667 ms window of EEG data starting from the moment the first flash happened. The x-axis was converted into milliseconds by dividing the sample index by the sampling frequency (240 Hz) to get the time in seconds and multiplying by 1000 to get ms.

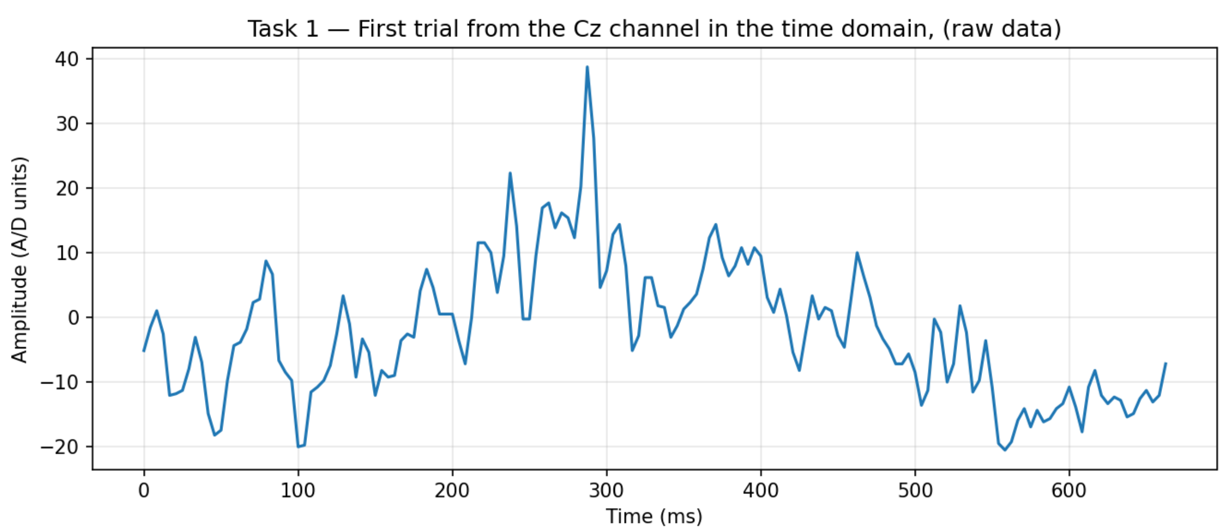


**Figure 1:** First trial from the Cz channel, raw EEG signal in the time domain.

The raw signal is quite noisy and has many rapid fluctuations all over the 667 ms window. There is a noticeable sharp spike around 290–300 ms that kind of stands out from the rest, which could be an early hint of the P300, but it is really hard to confirm because there is so much high-frequency noise mixed in with everything. This is basically what unprocessed EEG looks like and it is completely normal, but it also shows exactly why filtering is so important before doing any real analysis on the signal.

## Task 2 - FFT of the Raw Signal

Next, the Fast Fourier Transform (FFT) was applied to convert the signal from the time domain into the frequency domain. Instead of seeing voltage changing over time, the FFT shows how much of each frequency is present in the signal. The x-axis goes from 0 to 120 Hz, which is the Nyquist limit for a 240 Hz sampling rate, meaning we cannot detect anything above that. The FFT output is symmetric so the second half is just a mirror of the first, which is why only the positive frequencies are kept. We only care about how strong each frequency is, not its phase or sign.

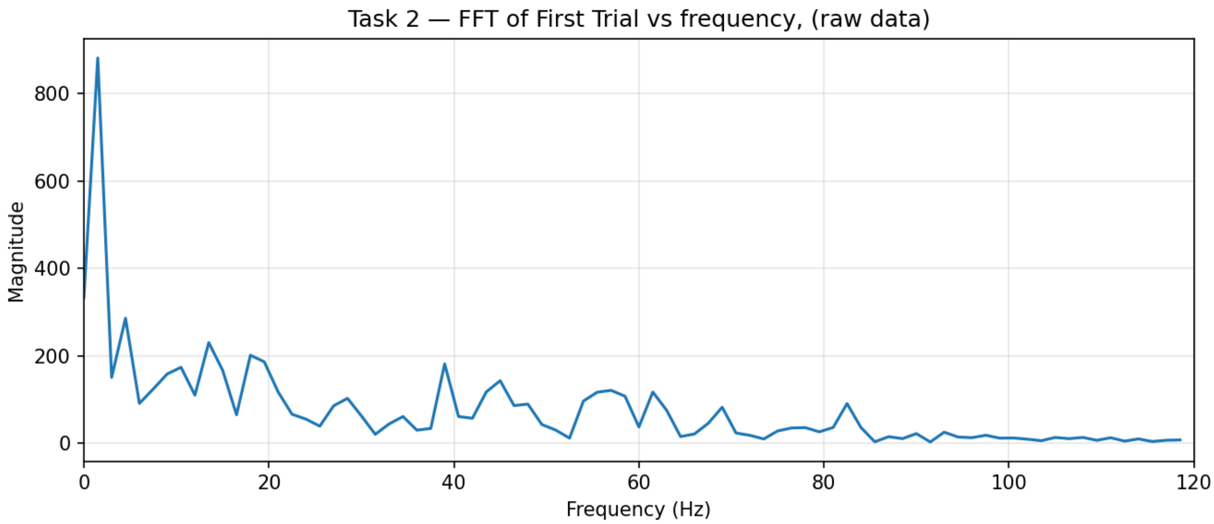


**Figure 2:** FFT magnitude spectrum of the first raw trial from the Cz channel.

The FFT of the raw signal shows energy spread over a wide range of frequencies, not just the low ones. The highest magnitudes are at low frequencies (around 0-5 Hz) which is expected because slow brainwave activity dominates EEG. But there are also significant contributions at higher frequencies, which are mostly coming from muscle artifacts, electrode noise, and other sources that are not related to the actual brain activity we want to see. This is exactly what tells us that filtering is needed to isolate the relevant signals.

## Task 3 - Detrending and Bandpass Filtering (0.5–8 Hz)

For this step, the signal was first detrended by subtracting the best-fit straight line from the whole epoch. This removes any slow linear drift in the signal, which can happen due to electrode movement or sweat. Then a 4th order Butterworth bandpass filter with cutoffs at 0.5 Hz and 8 Hz was applied. The P300 component lives in the low-frequency range, roughly 1–8 Hz, so this filter keeps what is actually relevant and removes everything else. The filtfilt function was used, which h applies the filter forward and backward so there is no time shift introduced in the signal.

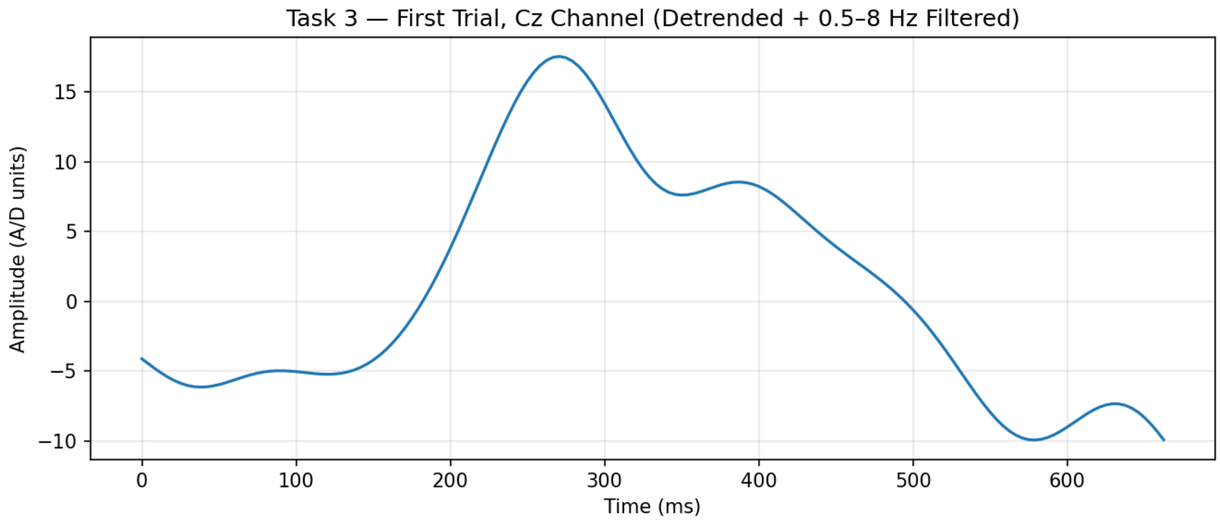


**Figure 3:** First trial after detrending and 0.5–8 Hz bandpass filtering, time domain.

The difference between the raw signal and this one is really surprising. The filtered signal is much smoother and the slow wave structure can be seen by now, there is a clear large positive peak around 270-300 ms that was completely hidden by the noise before. All those rapid fluctuations that were covering everything are gone, and what remains are the slow oscillations that actually contain meaningful and needed neural activity. Compared to Figure 1, this is a completely different picture and much easier to understand.

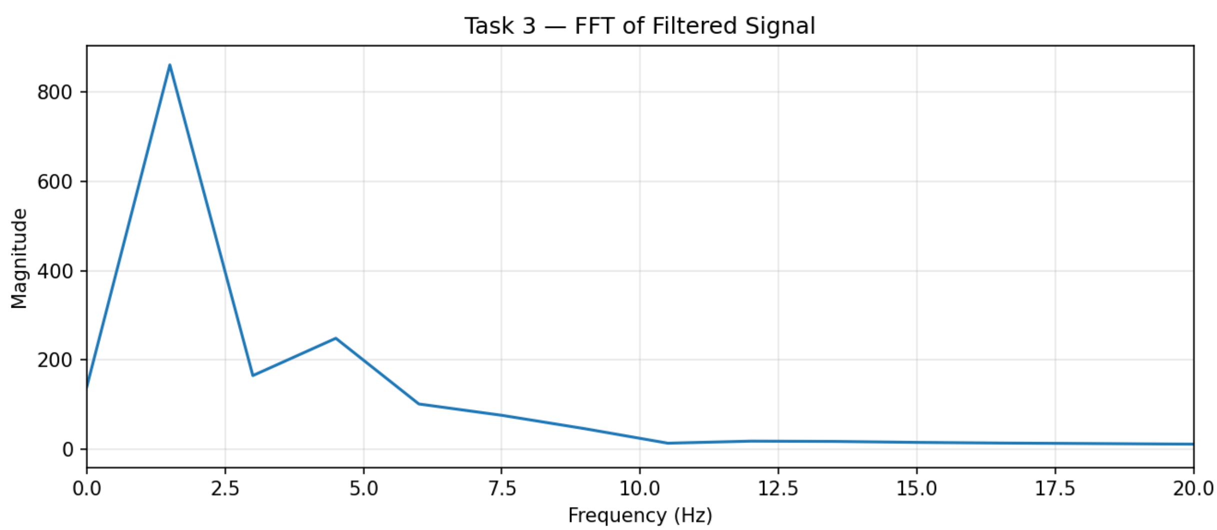


**Figure 4:** FFT magnitude spectrum of the filtered signal (zoomed to 0–20 Hz).

The FFT of the filtered signal shows that the filter worked exactly as expected. Almost all the energy above 8 Hz is now removed, and the spectrum is now concentrated in the low-frequency range with the main peaks below 8 Hz, and here exactly where we can find the needed p300. The signal gradually rolls off and becomes essentially flat after around 10 Hz. This actually makes sense because the Butterworth filter has a gradual rolloff, it doesn't cut everything above 8 Hz instantly, it just starts attenuating there and by 10 Hz it's basically gone. The x-axis was zoomed to 0-20 Hz here because there is basically nothing happening after that after filtering, and it makes the passband behavior much easier to see compared to the full 0-120 Hz range.

**Task 4 - Power Spectrum of the Filtered Signal**

The power spectrum was calculated by squaring the FFT magnitude values, so power = $|FFT|^2$. This shows the energy at each frequency instead of just the amplitude. The y-axis values here are in the millions ($1e^6$ scale) because squaring large amplitudes gives very large numbers.

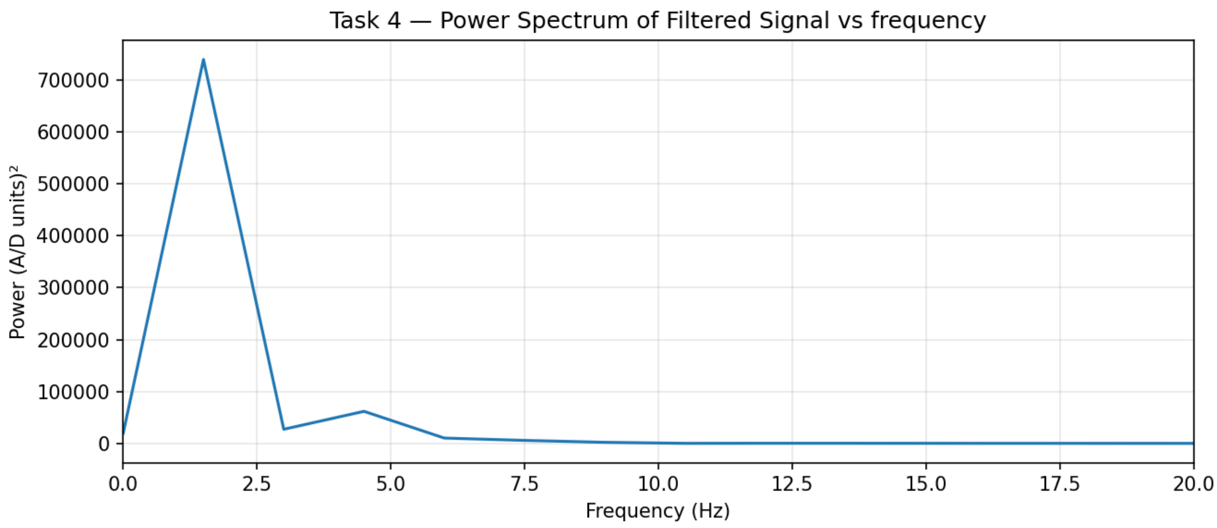


**Figure 5:** Power spectrum ($|FFT|^2$) of the filtered signal.

The power spectrum shows that the energy is very concentrated at the lowest frequencies, with a dominant peak near 1 Hz and another small one around 4 Hz. Above 8 Hz the power is flat and close to zero, which because the filter did its job. The differences between frequencies are much more exaggerated here than in the magnitude spectrum because of the squaring, which makes it harder to see the smaller peaks clearly. That is why looking at the log power in the next step is useful.

**Task 5 - Log Power Spectrum**

The logarithm of the power was computed using the formula 10 × log10(power), which converts the values to decibels (dB), which measures the intensity of a signal or the reduction of noise. This is useful because EEG power covers many orders of magnitude, and the linear scale makes everything except the biggest peak look flat. The log scale compresses the dynamic range so what is going on at lower power levels can be seen clearly.

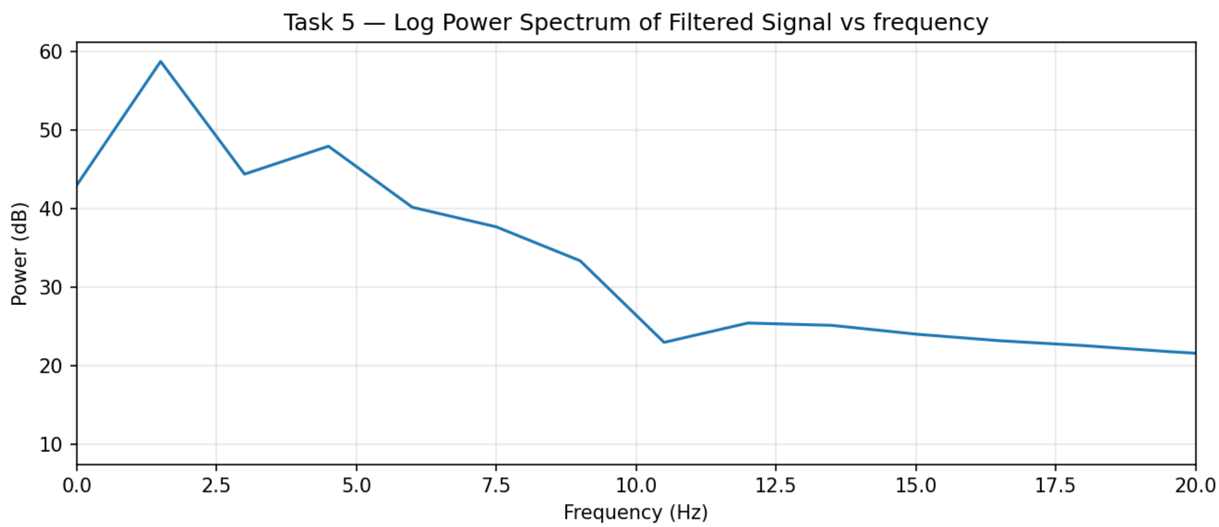


**Figure 6:** Log power spectrum (dB) of the filtered signal.

The log power plot is much more informative than the linear one. Now multiple peaks within about the 0-10 Hz and slightly beyond the passband, a dominant peak near 1 Hz and a clear one around 4–5 Hz, after which the power gradually decreases and reaches a dip around 10 Hz before flattening out, which is consistent with the Butterworth filter's rolloff effect. After 8 Hz the power decreases more gradually in the log domain, which makes sense since the Butterworth filter has a smooth rolloff rather than a perfectly sharp cutoff. The general trend of power decreasing with increasing frequency, known as the 1/f pattern, is also visible here and that is actually a well-known characteristic of EEG signals in general.

## <u>Task 6 - Average Target vs. Non-Target Responses</u>

This was the most important part of the assignment. For the first character epoch, all the individual flash responses were classified into two groups based on the StimulusType variable: flashes where the target row or column was lit up (StimulusType = 1, target) and all other flashes (StimulusType = 0, non-target). The 667 ms signal segment starting from each flash onset was taken, and then all segments within each group were averaged together. When you average many trials like this, the random noise cancels out because it is not consistent, but the P300 should survive the averaging since it always appears at the same time (~300 ms) for the target flashes. This was done separately for both the raw and filtered data.

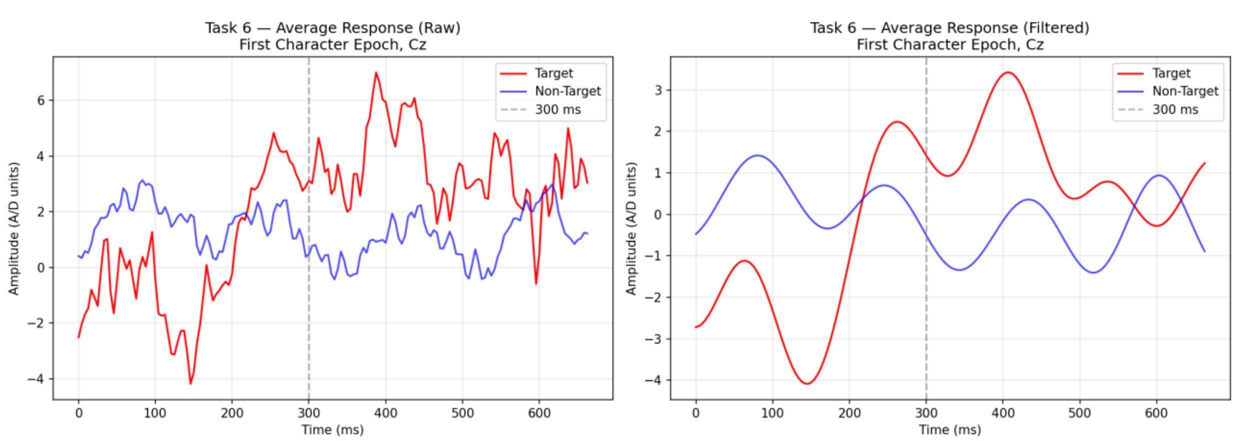

**Figure 7:** Average target (red) vs. non-target (blue) responses for the first character epoch, Cz channel. Raw on the left, filtered on the right. Dashed line marks 300 ms.

In the raw version on the left, the two curves are quite noisy and it is hard to clearly separate them by eye. They cross each other multiple times and there is no obvious consistent difference around the 300 ms mark, which is expected since the raw signal still has all the noise in it.

The filtered version on the right is where things get much clearer and more obvious. The red curve (target) shows a clear positive rise starting around 250 ms and reaching its peak around 400 ms, while the blue (non-target) curve does not show this same rise at all instead it stays on the same level of its oscillations and relatively low throughout. The peak appearing at nearly 400 ms rather than exactly 300 ms is completely normal, since the P300 component can appear anywhere between 250-500 ms depending on the task difficulty and the individual person. The "300 ms" label is just a general name for the component, not an exact fixed timing.

This difference between target and non-target is exactly what the P300 BCI system relies on to figure out which character the person was looking at, and the fact that it is basically invisible in the raw version but clearly visible here in the filtered one, really shows how crucial the filtering step is. Without the filter, the noise hides the actual brain response and the system would not be able to work accurately and any result would be said then is only built on guessing.

- ## Discussion

The aim of this assignment was to process EEG data from a P300 BCI paradigm and go through the main steps of signal analysis, starting from just plotting the raw signal all the way to comparing the averaged target and non-target responses. Looking at everything together across all the tasks, the processing pipeline worked as expected and each step made sense when looked at the previous one.

Starting with the raw signal in Task 1, it was obvious that the unprocessed EEG is just full of noise and rapid fluctuations that make it really hard to see anything meaningful. The FFT in Task 2 confirmed this, since the energy was spread across a really wide frequency range all the way up to 120 Hz, which should not be the case if it was only brain activity. This basically proved that filtering was necessary. After detrending and applying the 0.5-8 Hz Butterworth bandpass filter in Task 3, the signal became much cleaner and the slow wave structure actually became visible, which was not the case in the raw version. The FFT of the filtered signal also confirmed that the energy was now concentrated below 10 Hz, which is exactly where the P300 lives. The power spectrum in Task 4 showed the same thing but the differences were much higher because of the squaring in the formula, and the log power in Task 5 made it even easier to see the peaks within the passband. So each step was basically built on the previous one in a logical way.

The most important result was in Task 6. In the raw version, the difference between the target and non-target curves was barely visible because the noise was still overwhelming everything and the two curves were crossing each other all over the place. But in the filtered version, the target curve showed a really clear and strong positive peak around 400 ms that the non-target curve did not show at all, which is consistent with the expected P300 response. The peak appearing at 400 ms rather than exactly 300 ms is completely normal since the P300 timing varies between 250–500 ms depending on the person and task, specially since the FFT of the filtered signal showed that the energy was below 10 Hz not 8 Hz due to the Butterworth filter gradual rolloff. This difference is literally what the whole P300 BCI system relies on to figure out which character the person was looking at, so seeing it appear so clearly after filtering really confirmed that the preprocessing steps were not just necessary but actually really effective too.

However, there are some limitations should be mentioned. The analysis was only done on the first character epoch, which means the averages were calculated from a relatively small number of trials, and that could affect how clean the result is. Using more epochs and averaging across more characters would definitely give a stronger and cleaner P300 with less noise remaining. Also, since only a single trial was plotted in Tasks 1 and 3, those plots are specific to that one trial and its noise, so they might not perfectly represent the general signal behavior. In a more complete analysis, averaging multiple trials before plotting would be more reliable and accurate. But even with these limitations, the P300 effect was still clearly visible, which shows that the signal is strong enough to survive even with minimal averaging, and that the filter did its job well.

- **Conclusion**

Overall, the experiment successfully went through all the main steps of EEG signal processing and showed how the P300 response can be extracted from noisy raw data. Starting from a very messy raw signal that looked like pure noise, and ending up with a clean comparison where the P300 effect is actually visible in the filtered means, which was quite satisfying and interesting.

The most important thing was how much difference filtering the bandpass makes. The raw signal and the filtered signal look like completely different things, and without filtering, the comparison between target and non-target responses in Task 6 would be very hard to understand or explain. The FFT and power spectrum steps also helped understand why filtering was needed in the first place, since the raw FFT clearly showed energy spread across all frequencies, most of these noises and not brain activity.

One thing that could be improved is working with more epochs instead of just the first character. Averaging across more characters would give a stronger and cleaner P300 estimate. But even with just one epoch, the effect is visible and clear, which shows that the P300 has a strong and consistent response in this dataset. Python was used is this assignment, however Matlab could also be used giving the same results.

- **References:**

[1] M. Piela and M. P. Kotas, “Spatio-temporal matched filter adjustment for enhanced accuracy in brain responses classification,” *Journal of Applied Biomedicine*, vol. 45, no. 1, pp. 34–51, Dec. 2024, doi: 10.1016/j.bbe.2024.12.003.